

Notes on Real Analysis

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This set of notes shows my thoughts as I work through Rudin's Principles of Mathematical Analysis.

We begin from page 31.

A set $E \subset \mathbb{R}^k$ is convex if

$$\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in E$$

for $\mathbf{x}, \mathbf{y} \in E$ and $0 < \lambda < 1$.

Problem

Show that an open ball B in $\mathbb{R}^k$ with centre $\mathbf{c}$ and radius r is convex.

Solution

B is the set of $\mathbf{x} \in \mathbb{R}^k$ such that $|\mathbf{x} - \mathbf{c}| < r$. Suppose that $\mathbf{x}, \mathbf{y} \in B$. Let $0 < \lambda < 1$. We want to show that the vector $\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in B$. We want to show that

$$|[\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}] - \mathbf{c}| < r.$$

Consider

$$\begin{aligned} |[\lambda \mathbf{x} + (1 - \lambda)\mathbf{y}] - \mathbf{c}| &= |\lambda \mathbf{x} - \lambda \mathbf{c} + (1 - \lambda)\mathbf{y} + \lambda \mathbf{c} - \mathbf{c}| && \text{(Adding 0.)} \\ &= |\lambda(\mathbf{x} - \mathbf{c}) + (1 - \lambda)(\mathbf{y} - \mathbf{c})| \\ &\leq \lambda|\mathbf{x} - \mathbf{c}| + (1 - \lambda)|\mathbf{y} - \mathbf{c}| && \text{(Triangle ineq.)} \\ &< \lambda r + (1 - \lambda)r && \text{(Def. of a ball.)} \\ &= r. \end{aligned}$$

As desired. ■

Problem

Show that a k -cell, C in $\mathbb{R}^k$ is convex.

Solution

For two sequences $(a_i)_{i=1}^k, (b_i)_{i=1}^k$ with $a_i < b_i$, some vector $\mathbf{x} \in \mathbb{R}^k$ is in C if $\mathbf{x} := \langle x_1, x_2, \dots, x_k \rangle$ satisfy $a_i \leq x_i \leq b_i$ for $(i = 1, 2, \dots, k)$. Suppose that $\mathbf{x}, \mathbf{y} \in C$. We want to show that if $\mathbf{v} := \lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ for $0 < \lambda < 1$ then

$$a_i \leq v_i \leq b_i.$$

The i -th component of $\mathbf{v}$ is

$$\lambda x_i + (1 - \lambda) y_i$$

and because $\mathbf{x}, \mathbf{y} \in C$ they satisfy

$$\lambda a_i + (1 - \lambda) a_i \leq \lambda x_i + (1 - \lambda) y_i \leq \lambda b_i + (1 - \lambda) b_i$$

$$a_i \leq \lambda x_i + (1 - \lambda) y_i \leq b_i$$

and so $\mathbf{v}$ is in C , as desired. ■

Let X be a metric space. A *neighbourhood* of p , $N_r(p)$ is just the set of points of distance $< r$ from p , for some $r > 0$.

Some point p is a *limit point* of $E \subset X$ if every neighbourhood of p contains a point $q \neq p$ such that $q \in E$.